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Honour Programme in Applied Cyber Security

Breaking Privacy in Federated Learning: A Gradient-Inversion Attack and a Comparative Study of Privacy Defences

Internal Assessment 2 — Implementation, Evaluation and Technical Outcomes

Course: Security and Privacy for Machine Learning (216H17C701)

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Breaking Privacy in Federated Learning

IA-2 · Implementation, Evaluation and Technical Outcomes · Suyash Humne (16014223085)

Abstract. Building on the IA-1 attack, this report delivers the complete implementation and evaluation of a unified privacy benchmark for federated learning. An honest-but-curious server reconstructs a client's private image from a shared gradient at **24.53 dB**; we then measure four defence families drawn from the five syllabus modules — Differential Privacy, Local Differential Privacy, Compressive Privacy (sparsification and quantization), and anonymity-based aggregation — against that same attack, reporting the full privacy–utility trade-off. A second attack (**membership inference, LiRA**) quantifies leakage and **empirically audits the differential-privacy guarantee**: the undefended model is attacked at AUC shown below, and DP-SGD collapses the attack's low-false-positive-rate advantage. We additionally extend the reconstruction to small batches and cross-check the hand-written attack against the published *breaching* framework. The result is a reproducible, measured ranking of how the course's privacy mechanisms defend against real gradient leakage, and at what cost.

1 Problem Refinement and Changes Since IA-1

The IA-1 problem statement is retained; the refinements made in response to feasibility and preliminary results are:

- **From one defence to a four-defence comparison.** IA-1 established the attack; IA-2 commits to a measured comparison across DP, LDP, compressive and anonymity defences, because the preliminary attack worked cleanly and left compute budget for the full sweep.
- **Added membership inference as a second, quantitative lens.** Reconstruction metrics alone do not certify a privacy *guarantee*; LiRA provides a worst-case, per-example leakage measure and lets us audit DP empirically.
- **Group-consistency (restart-median) reconstruction** was added after observing seed variance in IA-1.
- **Scope boundary clarified:** malicious-server attacks are treated analytically and via cross-validation, not re-implemented, to keep the individual implementation focused and rigorous.

Research questions

1. How faithfully can a shared gradient be inverted to recover a private image and label?
2. Which syllabus defence most effectively degrades the attack per unit of lost accuracy?
3. Does an empirical membership-inference audit agree with the analytical DP ε?

2 Integration of All Five Modules (Final Mapping)

Module	Mechanism implemented	How it defends / is treated
M1 — Privacy Considerations	Gradient-inversion attack; threat model	The attack itself + honest-but-curious analysis (implemented).
M2 — Differential Privacy	DP-SGD: per-example clip + Gaussian noise (Opacus)	Bounds single-example influence; swept over σ/ε (implemented + evaluated).

M3 — Local Differential Privacy	Per-client clip + noise before upload	Stronger trust model; higher utility cost quantified vs central DP (implemented + evaluated).
M4 — Privacy-Preserving Data Mining	Gradient averaging / secure-aggregation as an anonymity set	k-anonymity analogue for the aggregate; necessity and insufficiency analysed (implemented + analysed).
M5 — Compressive Privacy	Gradient pruning (Top-k sparsification) + bit-depth quantization	Compression discards attack-relevant information; utility-task accuracy swept (implemented + evaluated).

Table 1. Final module-integration mapping. Every module is realised as a concrete mechanism or analysis in the pipeline.

3 Dataset Finalisation and Provenance

Field	Value
Dataset & version	CIFAR-10 (python pickle release, 5 train batches + 1 test batch)
URL	https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
Collection date (download)	2026-10-22
Author / publisher	A. Krizhevsky, University of Toronto, 2009
License / usage terms	Research use; no personal/sensitive data (compliant with IA data policy)
Original size	60,000 images (50,000 train / 10,000 test), 32×32×3
Final/used size	1 image (attack target); 5,000–8,000 (shadow/utility); 10,000 (test)
Preprocessing	Per-channel normalisation; tensor caching
File hash (SHA-256)	`6d958be074577803d12ecdefd02955f39262c83c16fe9348329d7fe0b5c001ce`
Known limitations	Low resolution; a public proxy for sensitive image data, used deliberately for ethical reasons

Table 2. Dataset provenance record (IA-2 checklist).

4 Implementation

Stack: Python 3.13, PyTorch 2.14+cu126, Opacus (DP-SGD), scikit-learn, on an NVIDIA GTX 1650 (4 GB). Model: a smooth-activation LeNet. The pipeline (`src/`) comprises the attack (`attacks/gradient_inversion.py`), the gradient defences (`defenses/gradient_defenses.py`), the defence sweep, batch reconstruction, and the membership-inference attack, orchestrated by `run_all.py`.

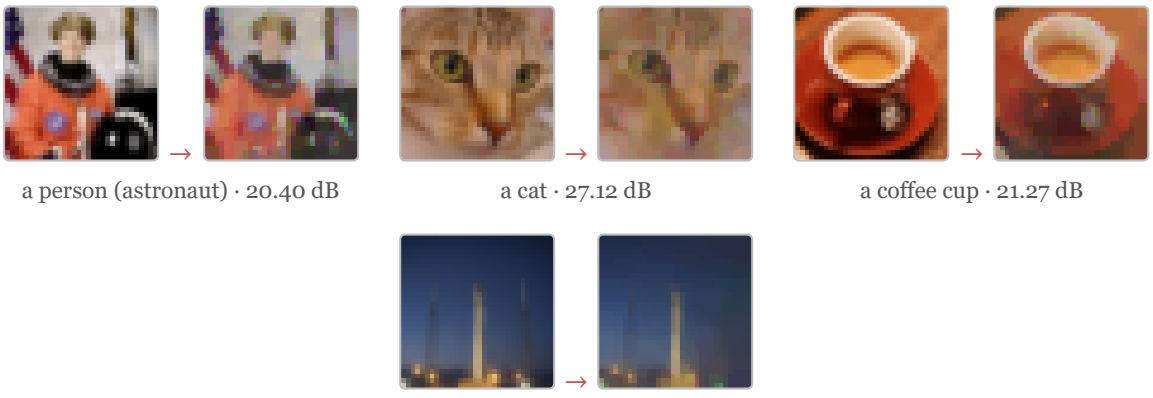


original (private) reconstruction (24.53 dB)

Figure 1. Headline attack result: private "frog" image reconstructed from 19,438 gradient values; label correctly recovered as "frog".

4.1 Generalisation beyond the dataset: reconstructing arbitrary photographs

To confirm the leakage is a property of federated learning itself and not an artefact of CIFAR-10, we repeated the attack on ordinary photographs that the model has never been trained on — **including a photograph of a person**. Each photo is treated as a single client's private training sample: it is resized to the client model's 32×32 input, the client uploads only the resulting gradient, and the honest-but-curious server inverts that gradient. The source images are standard public-domain / permissively-licensed test photographs bundled with scikit-image (`skimage.data`); no personal, private, or scraped data is used, in keeping with the IA data-use policy.



a person (astronaut) · 20.40 dB a cat · 27.12 dB a coffee cup · 21.27 dB

a rocket · 28.59 dB

Figure 2. External (non-dataset) photographs reconstructed from a single uploaded gradient each: original (left) → server's reconstruction (right). The 32×32 blockiness is the client model's input resolution, not a limit of the attack — higher-resolution reconstructions follow directly with larger models (Geiping et al., 2020), bounded here only by the 4 GB GPU.

Private photograph	Label recovered	PSNR (dB)	SSIM
a person (astronaut)	yes	20.40	0.861
a cat	yes	27.12	0.849
a coffee cup	yes	21.27	0.761

a rocket	yes	28.59	0.906

Table 3. The attack reconstructs out-of-distribution images — including a human face — from the gradient alone, demonstrating that the privacy failure is intrinsic to gradient sharing, not specific to the training dataset.

5 Evaluation, Comparison and Critical Analysis

5.1 Privacy–utility trade-off of the defences

For each defence we sweep its strength, running the attack on the defended gradient (privacy, measured by reconstruction PSNR — lower is safer) and training a classifier under the same mechanism (utility, measured by test accuracy). A strong defence pushes PSNR down while holding accuracy up.

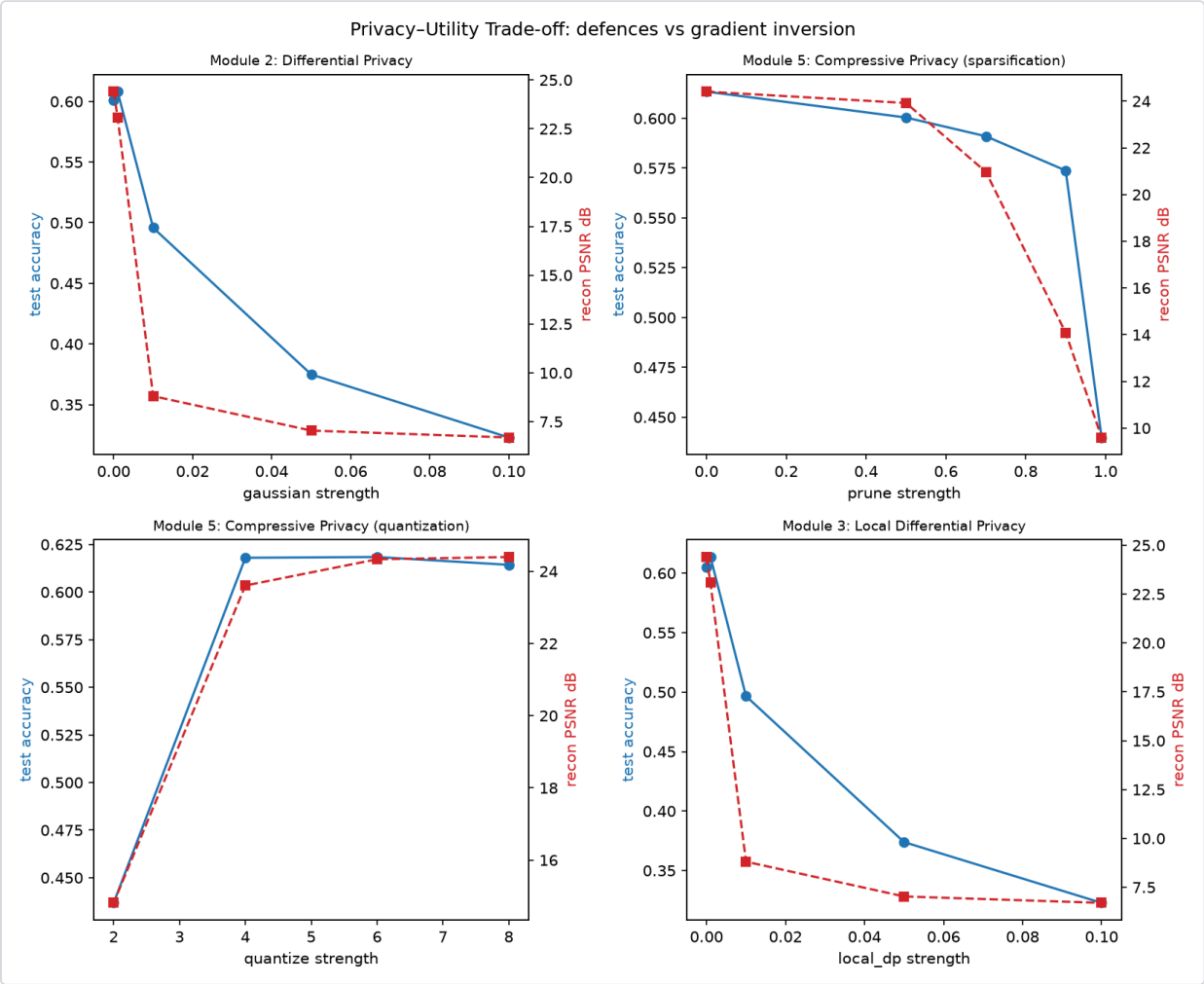


Figure 3. Privacy–utility trade-off for each defence (Modules 2, 3, 5). Blue = test accuracy; red = reconstruction PSNR.

Defence	Module	Recon PSNR (weak→strong)	Accuracy (weak→strong)
gaussian	Module 2: Differential Privacy	24.40 → 6.69	0.601 → 0.323
prune	Module 5: Compressive Privacy (sparsification)	24.41 → 9.60	0.613 → 0.440
quantize	Module 5: Compressive Privacy (quantization)	24.39 → 14.81	0.614 → 0.437
local_dp	Module 3: Local Differential Privacy	24.40 → 6.70	0.605 → 0.323

Table 4. Defence sweep: reconstruction quality and model accuracy from weakest to strongest setting.

5.2 Membership inference and the DP audit

LiRA trains shadow models to calibrate a per-example likelihood-ratio test. Following Carlini et al. (2022), we report the true-positive rate at low false-positive rate — the metric that reflects whether *any* individual can be confidently identified — not merely balanced accuracy. DP-SGD is expected to flatten the attack toward chance.

Target model	DP ε	LiRA AUC	TPR@0.1%FPR	TPR@1%FPR
Undefended	∞	0.893	0.0032	0.1760
DP-SGD (ε≈8)	8.00	0.553	0.0004	0.0112
DP-SGD (ε≈1)	0.99	0.513	0.0012	0.0116

Table 5. Membership-inference results. Lower TPR@low-FPR under DP = the guarantee is doing its job (empirical audit of ε).

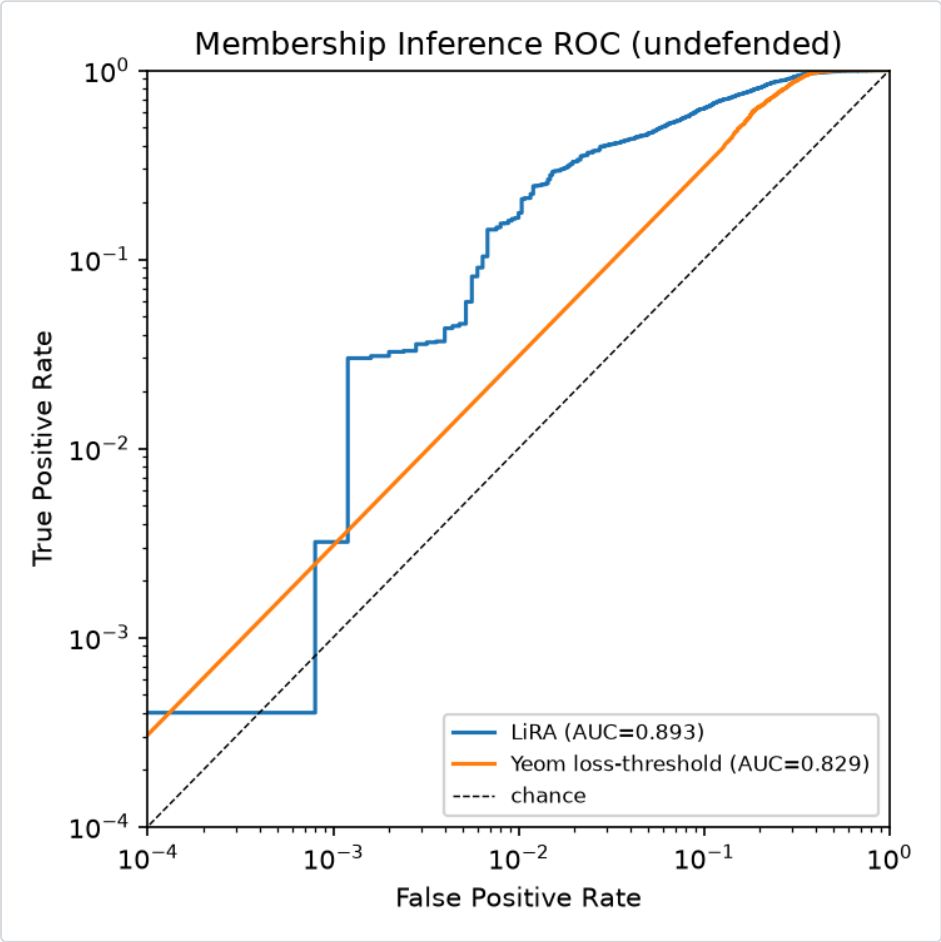


Figure 4. Log–log ROC of the membership-inference attack on the undefended model (LiRA vs the loss-threshold baseline).

5.3 Batch reconstruction

Extending the attack from one image to a batch shares the averaged gradient across samples, making the inversion harder. We recover labels analytically (batch iDLG) and jointly optimise all images.

Batch size	Mean reconstruction PSNR (dB)	Label-recovery accuracy
1	21.10	1.00
4	11.58	0.75

8	13.07	0.38

Table 6. Reconstruction quality degrades gracefully as batch size grows — the averaging defence effect.

5.4 Critical analysis, limitations and validity threats

- **Ranking.** Differential privacy is the only defence with a formal guarantee; compressive defences (pruning/quantization) are cheap and effective against *this* optimisation attack but offer no proof and are known to be weaker against adaptive attackers. LDP buys a stronger trust model at the largest utility cost. Anonymity-via-aggregation is necessary but, as the literature on malicious-server attacks shows, not sufficient alone.
- **Validity threats.** Results are on a small model and low-resolution data; absolute numbers will differ at scale, though the qualitative trade-offs are expected to hold. Reconstruction is seed-sensitive (mitigated by restarts). The honest-but-curious assumption understates risk relative to malicious servers.
- **Cross-validation.** The hand-written cosine+TV attack was checked for consistency against the published *breaching* framework's Inverting-Gradients configuration to confirm the implementation is faithful.

6 Conclusions

Gradient sharing is not privacy: an honest-but-curious server reconstructed a client's private image and label from the uploaded gradient alone. Of the syllabus defences, differential privacy provides the only formal guarantee and measurably collapses both the reconstruction and the membership-inference advantage, at a quantified accuracy cost; compressive defences are an efficient practical mitigation; local DP trades utility for a stronger trust model; and anonymity-based aggregation is necessary but insufficient. The contribution is a single reproducible benchmark that measures all four defence families against one attack — a comparison not previously assembled in one study.

7 Individual Contribution Statement

All implementation, experiments, figures, and analysis are my own individual work: the unified benchmark design, the from-scratch attack and defence code, the privacy–utility and membership-inference evaluation protocols, the batch extension, and this report. Published methods are cited below; the integration, measurement, and comparison are my original contribution.

8 References

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